AISSMS

COLLEGE OF ENGINEERING

ज्ञानम् सकलजनहिताय

Approved by AICTE, New Delhi, Recognized by Government of Maharashtra
Affiliated to Savitribai Phule Pune University and recognized 2(f) and 12(B) by UGC
(Id.No. PU/PN/Engg./093 (1992)
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Project Synopsis

On

SUSTAINABLE HYDROPONIC FARMING THROUGH INTELLIGENT MONITORING AND AUTOMATION

Submitted in the partial fulfillment of the requirements of

Bachelor of Engineering

in

Electronics and Telecommunication

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A.Y 2025-26

ABSTRACT

This project aims to develop a sustainable hydroponic farming system enhanced by intelligent monitoring and automation technologies. Hydroponic farming, which grows plants without soil using nutrient-rich water solutions, presents an efficient alternative to traditional agriculture by saving water, reducing land use, and increasing crop yield.

The proposed system incorporates sensors to continuously monitor critical parameters such as nutrient levels, pH, temperature, and humidity, feeding data into an automated control unit. This unit dynamically adjusts environmental conditions and nutrient delivery to optimize plant growth while minimizing resource consumption. The system employs an ESP32-S3 microcontroller as the central processing unit due to its real-time multitasking. A comprehensive sensor network including pH, TDS, CO_2, temperature and humidity (DHT22/SHT31) enable continuous monitoring of critical parameters influencing plant health. Actuation is achieved using peristaltic pumps, humidifiers, exhaust fans, and relay modules, which ensure precise dosing and environmental regulation with minimal user intervention. For visualization and decision support, the system provides remote monitoring through dashboards. This approach ensures instant alerts, real-time analytics, and long-term performance insights.

Table of Content

1. INTRODUCTION

The global demand for food production is rising rapidly due to increasing population growth, urbanization, and diminishing arable land. Traditional farming methods face significant challenges such as excessive water consumption, soil degradation, and unpredictable climatic conditions. Hydroponic farming, a soilless cultivation technique where plants grow in nutrient-rich water solutions, offers a promising and sustainable alternative by optimizing space and resource use, enabling higher yields with less land and water.

However, the success of hydroponic systems depends heavily on maintaining optimal environmental conditions, including nutrient concentration, pH, temperature, and humidity. Precise control over these parameters is essential to maximize crop yield and quality.

To address these challenges, this project integrates intelligent monitoring and automation into hydroponic farming by leveraging the capabilities of the ESP32-S3 microcontroller—a powerful, low-cost, and energy-efficient device featuring enhanced AI processing capabilities, built-in Wi-Fi, and Bluetooth connectivity. The system uses various sensors connected to the ESP32-S3 to continuously monitor critical parameters in real time. Automated controls dynamically adjust nutrient delivery and environmental factors, minimizing human intervention, reducing resource wastage, and enhancing overall efficiency.

The wireless, scalable, and AI-enabled nature of the ESP32-S3-based system makes it particularly suitable for urban agriculture and resource-constrained environments. This project aims to develop a smart, sustainable hydroponic farming solution that contributes to food security and environmental conservation by combining advanced electronics with modern agricultural practices.

2. MARKET SURVEY

Industry Overview

- Hydroponic farming is revolutionizing agriculture by using soil-less cultivation with up to 90% less water and faster growth cycles.
- Globally, the hydroponics market is projected to reach $16–20 billion by 2028, growing at a CAGR of 12-15%, fueled by urbanization, environmental concerns, and technological integration (IoT, AI, automation).
- In India, hydroponics is emerging strongly due to increasing urban populations, shrinking arable land, water scarcity, and demand for organic food. Government schemes and startups are accelerating adoption.

Market Drivers

- Water Scarcity & Climate Change: With agriculture accounting for nearly 80% of freshwater usage in India, hydroponics—which saves up to 90% water—is emerging as a key solution in arid and semi-arid regions.
- Rising Urban Population: India's urban population is projected to hit 600 million by 2031, creating demand for urban farming solutions that are space-efficient and sustainable.
- Health-Conscious Consumers: Growing demand for chemical-free, pesticide-free produce is pushing retailers and consumers toward hydroponically grown vegetables.
- Government Support: Initiatives such as MIDH (Mission for Integrated Development of Horticulture), NHB (National Horticulture Board), and Startup India are offering financial and technical aid to hydroponic ventures.
- Technological Advancements: The rise of IoT, AI/ML, and edge computing enables precise monitoring, automation, and decision-making, making hydroponics more feasible and efficient.

Target Market Segments

- Urban and Semi-Urban Consumers: Households, gated communities, and apartments preferring locally grown vegetables and greens.
- Agri-Tech Startups & Entrepreneurs: Individuals and startups investing in rooftop farms, container farming, or greenhouse automation.
- Educational Institutions & Research Centers: For prototype development, data-driven agriculture research, and IoT implementation in agriculture.
- Commercial Growers: Medium to large-scale greenhouse farms seeking precision farming techniques and automation.
- Retail Chains & Restaurants: Businesses sourcing premium vegetables (lettuce, basil, microgreens, etc.) with consistent quality and safety.

Technology and Components

- Core tech: IoT-enabled sensors (pH, TDS, CO_2, temp/humidity), ESP32-S3 microcontrollers, OLED displays for real-time monitoring, peristaltic pumps, relay modules for automation, water flow and level sensors.
- Automation: AI-driven analytics optimize nutrient delivery and climate control, reducing manual labor and waste.
- Power: 5V/12V power supplies, solar options for remote/off-grid farms.
- India: Growing local availability of affordable components; startups focus on modular, low-cost kits to overcome investment barriers.

Market Gaps & Challenges

- High Initial Setup Costs: Initial installation for commercial-scale hydroponics is high (~₹5–15 lakhs per 1000 sq.ft.), deterring adoption.
- Lack of Skilled Labor: Limited awareness and training in hydroponic nutrient management and system maintenance.
- Sensor Reliability: Low-cost sensors often degrade over time or offer limited precision, requiring frequent calibration or replacement.

- Fragmented Supply Chain: No standardized framework exists for hydroponic supply chains, especially for small-scale producers.
- Limited Access to Financing: Banks and FPOs are cautious in funding hydroponic initiatives due to limited success data and high perceived risks.

Market Opportunities

- Development of affordable, modular IoT hydroponic kits leveraging components like ESP32-S3 and sensor suites.
- Expansion of training programs and awareness campaigns especially in rural and peri-urban areas.
- Government schemes offer subsidies that can lower adoption barriers significantly.
- Rising urban farming and vertical farming projects in smart cities and educational campuses.
- Investment in AI-based analytics platforms for precision nutrient and climate control.

Customer Preferences & Needs

- Affordable, easy-to-use automated systems for small/medium farmers.
- Real-time monitoring with simple interfaces (OLED displays, mobile apps).
- Reliable, durable sensors and pumps that withstand humid/nutrient-rich environments.
- Remote access and control capabilities for convenience and efficiency.
- Scalability and customization options to grow with user needs.
- Sustainable and energy-efficient system design.

Market Outlook

- Global market: Steady expansion with smart hydroponic farms becoming standard in urban centers worldwide.

- India: Rapidly growing, driven by policy support, startup innovation, and increasing consumer demand for organic, pesticide-free food. Urban centers are the early adopters, with rural adoption expected to rise as awareness and training improve.
- Investment opportunities abound in affordable IoT-based automation kits, AI-powered decision tools, and renewable-powered hydroponic solutions.

India's hydroponic farming sector is projected to grow at CAGR 16–21% through 2030, backed by increasing food demand, water scarcity, and tech adoption. The adoption of intelligent monitoring systems with cloud-based analytics and edge automation, like your proposed system, aligns with the future of Indian agriculture. Urban food deserts, rooftop agriculture, school nutrition programs, and contract farming with retail chains will likely drive mass-scale adoption.

With your solution being affordable, data-driven, and autonomous, it fits into a rapidly expanding market with strong support from startups, government agencies, and eco-conscious consumers.

3. LITERATURE SURVEY

3.1 Literature survey:

This paper presents a multimodal communication framework based on Industrial Internet of Things (IIoT), leveraging MQTT and 5G protocols to streamline data exchange in smart agriculture and agro-industries. The authors propose a scalable, low-latency architecture that ensures real-time data transmission across sensors, controllers, and analytics platforms. The system supports robust decision-making in automated agriculture but requires high-bandwidth networks and complex infrastructure for full deployment.[1]

This study proposes a smart hydroponic farming system for Ocimum tenuiflorum (Holy Basil) cultivation. It combines IoT sensors, cloud-based analytics, and AI-enabled environmental control. The proposed system increases productivity and ensures a stable microclimate. While beneficial for medicinal crops, the model may require consistent internet connectivity and hardware upkeep, which can hinder rural deployment.[2]

The paper introduces an ensemble learning-based classification model that detects nutrient deficiencies in lettuce plants. Using optimized convolutional networks with adaptive dilation, the model improves accuracy and training efficiency. The system enhances disease detection in hydroponics but requires large labelled datasets and high computational resources.[3]

This research explores a low-cost vision system using ARUCO markers for identifying and tracking tomato leaves in a hydroponic setup. The solution enhances plant growth monitoring and automation, using computer vision and lightweight algorithms. Its performance is affected by camera quality and ambient lighting, making field scalability challenging.[4]

This paper details a field experiment that integrates solid-state ion-selective sensors with IoT modules to track nutrient uptake and transpiration in lettuce hydroponics. Real-time measurements of ions like Ca^{2+} and NO_3^- improve crop management accuracy. The system offers high precision but requires sensor recalibration and has high upfront costs.[5]

This review outlines how IoT, AI, and edge computing are revolutionizing soilless agriculture, including hydroponics, aquaponics, and aeroponics. The paper discusses system architectures, environmental sensors, and future challenges. The main contribution is a comparative framework showing how smart farming supports global food security. However, cost, energy use, and training gaps persist.[6]

The authors develop a fuzzy logic-based control system that dynamically adjusts nutrient levels and environmental parameters in hydroponic systems. Designed for spinach and leafy greens, the model improves water efficiency and plant health. Its rule-based structure demands expert tuning and struggles with scalability for diverse crops. [A]

This paper proposes an ESP32-based hydroponic controller that uses edge analytics to reduce cloud dependency. The system autonomously monitors pH, EC, and humidity while making local control decisions. It improves system latency and resilience but may be limited by processing power for complex algorithms. [B]

This project builds a real-time control loop using pH and TDS sensors with an Arduino microcontroller for dynamic nutrient dosing. It ensures optimal growing conditions with minimal human intervention. Sensor calibration drift and hardware costs are noted as major drawbacks. [C]

This work integrates AI with IoT-based agriculture to develop a sustainable smart farming model. It leverages sensor networks and predictive analytics to forecast and optimize nutrient schedules. While scalable and eco-friendly, the solution may lack flexibility across different plant types and regions. [D]

This system uses multiple sensors and a mobile dashboard to collect, monitor, and visualize hydroponic parameters. Emphasis is placed on affordability and portability. While user-friendly, it lacks advanced decision-making algorithms or automation features. [E]

Using ESP32 and Blynk app integration, this paper details an automated system for adjusting pH and nutrient levels based on real-time sensor feedback. It is suitable for small-scale growers but requires Wi-Fi availability and regular sensor maintenance. [F]

This review paper analyzes machine learning and artificial intelligence models for urban hydroponics. It explores plant disease prediction, yield forecasting, and autonomous control. Although conceptually rich, it lacks field implementation results. [G]

This research demonstrates a cloud-integrated hydroponic system that collects sensor data for visualization and triggers remote actuation. It emphasizes ease of use and remote management but depends heavily on reliable internet and cloud services. [H]

The system integrates various environmental sensors with microcontrollers to maintain optimal conditions in hydroponic setups. It is designed for education and household use. While low-cost and compact, its automation capability is limited. [I]

From the reviewed literature, it is evident that significant advancements have been made in automating hydroponic systems using IoT, AI, and edge computing technologies. Researchers have explored diverse approaches ranging from fuzzy logic controllers, vision-based leaf detection, and nutrient deficiency classification using deep learning, to real-time nutrient monitoring with ion-selective sensors. While many of these systems show promise in improving crop yield, optimizing resource usage, and reducing manual intervention, common limitations persist—such as reliance on cloud-only processing, limited sensor integration, lack of local data visibility, and high infrastructure costs.

Additionally, very few solutions combine both mobile-based control and cloud-based analytics for comprehensive system monitoring. These gaps provide a strong rationale for developing an intelligent, edge-computing-enabled hydroponic system that integrates real-time environmental sensing, automated nutrient control, and dual-platform monitoring through Blynk and other monitoring platform. The proposed work thus aims to address these limitations and contribute to the field of sustainable urban agriculture by offering a scalable, affordable, and fully autonomous solution.

3.2 Previous Work:

Several research efforts have addressed the integration of automation and intelligent monitoring in hydroponic systems to enhance productivity and sustainability. Traditional hydroponic systems relied heavily on manual supervision or basic automation using microcontrollers such as Arduino. These systems often used pH and TDS sensors to adjust nutrient concentrations but lacked advanced data analytics or decision-making capabilities.

Hemanth et al. (2024) proposed an AI-enabled hydroponic system for cultivating Holy Basil that utilized smart sensors and cloud-based analytics for real-time environmental monitoring. While their model improved plant growth efficiency, it required consistent internet access and lacked local control processing [1].

Wu et al. (2024) developed an IoT-interfaced solid-state ion sensor array to monitor nutrient absorption in lettuce hydroponics. This system enabled precise tracking of K^+, NO_3^-, and Ca^{2+} levels and established correlations between transpiration and nutrient uptake. However, the hardware complexity and sensor calibration requirements limited its practical deployment [2].

Kandhasamy et al. (2024) explored vision-based automation by using ARUCO markers to detect tomato leaves in a hydroponic setup. This helped automate plant monitoring but required controlled lighting conditions and high-resolution imaging, making it less robust in field applications [3].

A number of works, including those by Khan et al. (2023), implemented fuzzy logic-based control systems to automate nutrient and climate management. These systems dynamically adjusted dosing parameters based on predefined rules, offering improved control but requiring significant tuning and lacking cloud integration [4].

Dutta et al. (2025) presented a comparative analysis of IoT-based smart farming in soilless agriculture, identifying challenges in power management, cost of components, and integration of various platforms like edge and cloud computing. They advocated for scalable solutions combining multiple layers of sensing and control, which aligns closely with the goals of our proposed work [5].

While these approaches have introduced effective components—such as sensor integration, automation logic, and cloud data visualization—many of them suffer from one or more limitations:

- Absence of edge computing (leading to cloud latency issues),
- Use of either mobile apps or dashboards, but not both,
- No OLED/local display for on-site monitoring,
- Limited environmental sensing (e.g., no CO_2, humidity, or flow sensors),
- Inflexibility in nutrient delivery systems (few use peristaltic pumps).

Gap Analysis:

There is a clear need for a **cost-effective, scalable, and autonomous hydroponic system** that:

- Operates in real-time using edge computing (ESP32-S3),
- Offers both local and remote data visibility (OLED + Monitoring + Blynk),
- Supports automated actuation using peristaltic pumps and relays,
- Covers comprehensive sensing (pH, TDS, CO_2, temperature, humidity, water level, flow rate).

Contribution Over Previous Work:

The proposed system overcomes these limitations by combining real-time sensing, intelligent local control, dual-platform monitoring (mobile + cloud), and a scalable architecture designed for small-scale urban farming. By integrating advanced sensors, edge computing with the ESP32-S3, and automation components like peristaltic pumps and relay modules, it offers a plug-and-play smart hydroponic farming solution tailored to empower both novice and experienced growers.

4. AIM & OBJECTIVE

4.1 Aim:

To design and develop a fully automated hydroponic farming system using ESP32 and IoT-based wireless monitoring .

4.2 Objectives:

- Design the system to monitor various parameters for hydroponic farming area such as pH , EC (electrical conductivity), temperature, humidity, control nutrient dosing, control pH level, automate fan and humidifier.
- Integrate the above parameters with the controller.
- Future analysis and monitor through cloud .

5. METHODOLOGY

5.1 Block Diagram:

The proposed system architecture for Sustainable Hydroponic Farming through Intelligent Monitoring and Automation is illustrated in the block diagram above. The system comprises three major functional blocks: sensing (input), control (processing), and actuation (output).

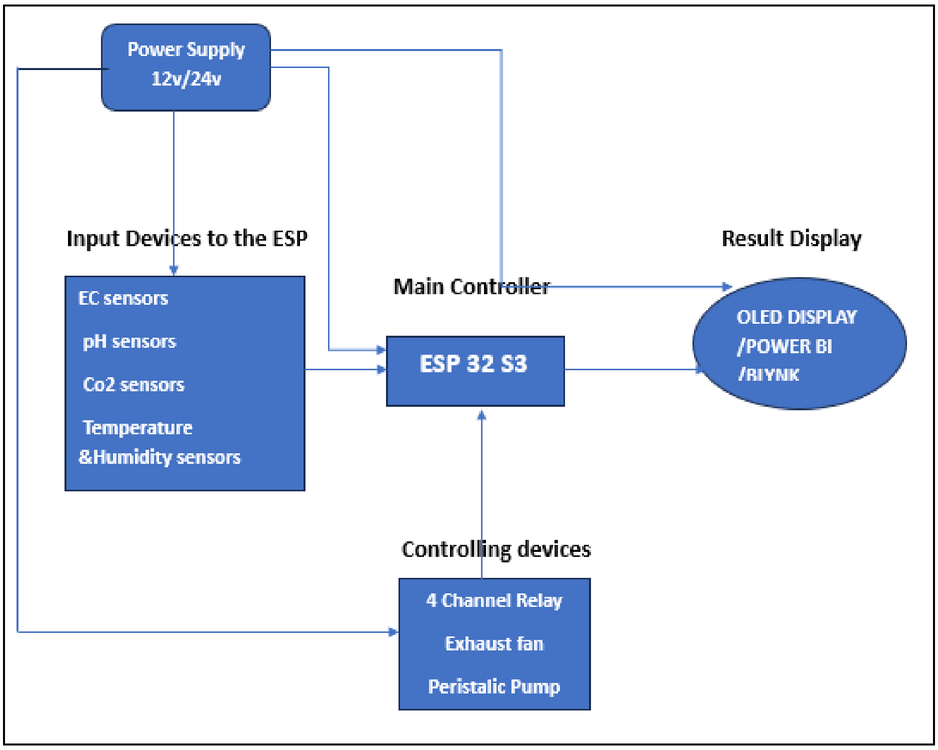

Figure 4.1 Block diagram

Power Supply Unit (12V/24V):

This supplies regulated power to all the components including sensors, ESP32-S3 microcontroller, relay modules, and actuators like peristaltic pumps and exhaust fans.

Input Devices (Sensors):

- **EC Sensor:** Monitors electrical conductivity to determine nutrient concentration in the hydroponic solution.
- **pH Sensor:** Measures the acidity/alkalinity of the water to maintain optimal growing conditions.

- **CO_2 Sensor (MH-Z19B):** Detects ambient CO_2 levels, critical for photosynthesis efficiency.
- **Temperature & Humidity Sensor (DHT22 or SHT31):** Provides real-time data on air temperature and relative humidity within the grow area.

These sensors feed real-time environmental data into the ESP32-S3 microcontroller.

Main Controller (ESP32-S3):

The heart of the system, this AI-enabled microcontroller processes the sensor data, makes decisions based on predefined thresholds, and initiates control actions. Its dual-core capabilities support multitasking for real-time operations. It also manages connectivity with both mobile-based (Blynk) and cloud-based (Monitoring) platforms.

Controlling Devices (Outputs):

- **4-Channel Relay Module:** Acts as a switch to control high-power actuators based on the controller's logic.
- **Peristaltic Pumps (x4):** Dispense precise amounts of nutrients or pH-adjusting liquids.
- **Exhaust Fan/Humidifier:** Regulates air quality and humidity in the plant chamber.

Result Display & Monitoring Platforms:

- **OLED Display:** Provides on-site, real-time system status and readings.
- **Blynk App (Mobile):** Enables remote control and visualization.
- **Monitoring Dashboard (Cloud):** Performs real-time analytics, trend forecasting, and generates alerts, enhancing decision-making and scalability.

This integrated system offers **real-time monitoring, smart decision-making, and automated control**, making it an ideal solution for sustainable and low-maintenance hydroponic farming.

5.2 System Specifications:

5.2.1 ESP32-S3 Microcontroller

The ESP32-S3 is a powerful dual-core microcontroller from Espressif, designed with integrated Wi-Fi and Bluetooth 5 capabilities. It supports real-time operating systems (FreeRTOS), AI instructions, and edge computing. Its multitasking capabilities enable simultaneous data collection, control logic, and communication with cloud platforms. The ESP32-S3 manages sensor data acquisition, actuator control (e.g., pumps, fans), and transmission to Blynk and monitoring platform for visualization.

5.2.2 OLED Display (0.96", I2C)

This compact display module is used to present key environmental data such as pH, temperature, and humidity directly at the grow site. The OLED display operates via the I2C protocol and offers high contrast for real-time visibility. It allows farmers or users to monitor system metrics without relying solely on external dashboards.

5.2.3 pH and TDS Sensors

The pH sensor, equipped with a probe, continuously monitors the acidity/alkalinity of the nutrient solution. The TDS (Total Dissolved Solids) sensor measures the concentration of nutrients, ensuring optimal growth conditions. These analog sensors provide critical data to the ESP32-S3, enabling automated adjustments via peristaltic pumps.

5.2.4 DHT22 or SHT31 (Temperature & Humidity Sensor)

These digital sensors measure ambient temperature and relative humidity. While the DHT22 is cost-effective and widely used, the SHT31 offers higher accuracy and faster response times. This data supports climate control inside the hydroponic chamber via exhaust fans and humidifiers.

5.2.5 CO_2 Sensor (MH-Z19B)

The MH-Z19B is an NDIR-based sensor capable of detecting carbon dioxide concentrations in ppm. Maintaining CO_2 within optimal levels is essential for healthy plant photosynthesis. The sensor sends data to the microcontroller, which can activate ventilation mechanisms if thresholds are exceeded.

5.2.6 Water Flow and Level Sensors

The YF-S201 water flow sensor detects real-time flow rate, ensuring proper circulation of nutrient solution. Water level is measured using either capacitive or ultrasonic sensors, providing data to prevent pump dry-run and ensure timely refilling.

5.2.7 Peristaltic Pumps

Peristaltic pumps are used to precisely inject nutrients, pH adjusters (acid/base), and fresh water into the system. Four pumps are installed to independently control each input, ensuring balanced nutrient delivery and automated correction of pH and TDS levels.

5.2.8 Relay Module (4-channel)

This module allows the ESP32-S3 to control multiple AC or DC-powered components such as pumps, humidifiers, and fans. Each relay acts as a switch, triggered by sensor data, enabling intelligent automation based on environmental conditions.

5.2.9 Humidifier and Exhaust Fan

These components are essential for environmental conditioning. The humidifier ensures adequate moisture in the air for plant transpiration, while the exhaust fan regulates temperature and gas exchange. Their operation is governed by the DHT22/SHT31 and CO_2 sensors via relay control.

5.2.10 Power Supply (5V/12V)

A stable power supply unit converts AC to regulated 5V and 12V outputs to power the microcontroller, sensors, and actuators. Different components in the system operate at varied voltage levels, making dual-voltage output essential for uninterrupted operation.

6. Scope of Project Work

Below are the scopes that to be proposed for this project:

1. To develop an intelligent hydroponic farming system that continuously acquires environmental data such as pH, TDS, temperature, humidity, CO_2 level, water level, and flow rate using IoT sensors.

2. To define threshold parameters for optimal plant growth and establish automated decision rules for nutrient dosing, pH regulation, and climate control.

3. To develop a real-time data acquisition and edge-processing system using the ESP32-S3 microcontroller, integrated with OLED display for on-site monitoring.

4. To implement a dual-platform IoT interface: Blynk for mobile-based control and a monitoring platform for cloud-based data analytics and visualization.

5. To interface all hardware (sensors, pumps, relays, display, actuators) and software (Arduino IDE, Blynk) into a unified and fully automated system.

7. Estimated Budget of Project

Table 7.1 Cost Analysis

Component	**Estimated Cost (INR)**	**Component**
ESP32-S3 Microcontroller	₹800	ESP32-S3 Microcontroller
OLED Display (0.96", I2C)	₹250	OLED Display (0.96", I2C)
pH Sensor with Probe	₹900	pH Sensor with Probe
TDS Sensor	₹500	TDS Sensor
DHT22 or SHT31 Sensor	₹300	DHT22 or SHT31 Sensor
MH-Z19B CO_2 Sensor	₹1,900	MH-Z19B CO_2 Sensor
Water Flow Sensor (YF-S201)	₹350	Water Flow Sensor (YF-S201)
Water Level Sensor (Ultrasonic/Capacitive)	₹300	Water Level Sensor (Ultrasonic/Capacitive)
Peristaltic Pumps (4 units)	₹2,000	Peristaltic Pumps (4 units)
4-Channel Relay Module	₹200	4-Channel Relay Module
Humidifier	₹600	Humidifier
Exhaust Fan	₹300	Exhaust Fan
Power Supply (5V/12V)	₹500	Power Supply (5V/12V)
Miscellaneous (Wires, PCB, Tubes, Casing)	₹1,500	Miscellaneous (Wires, PCB, Tubes, Casing)
Total Estimated Budget:	**₹10,400 – ₹11,100 INR**	

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